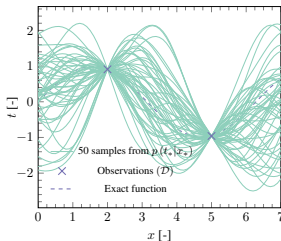


Bias through epistemic uncertainty

We again discard the micromodel, but more carefully

- Probabilistic surrogates, acknowledging we cannot compute σ everywhere

$$p(t|x) = \mathcal{N}(y(x, \mathbf{w}), \beta) \quad \Rightarrow \quad \sigma = \hat{\sigma}_{\mathbf{w}}(\varepsilon, \alpha) + \text{noise}$$



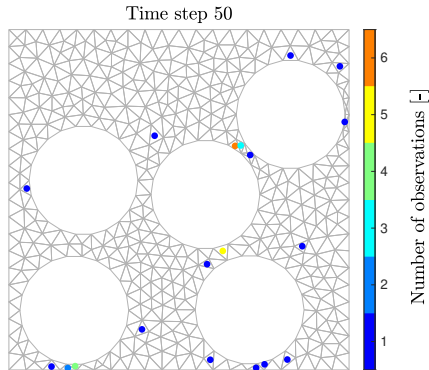
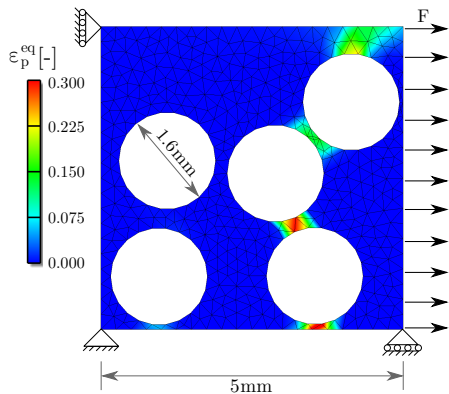
$$p(\mathbf{t}) = \mathcal{N}(\mathbf{t}|\mathbf{0}, \mathbf{K})$$

$$K_{pq} = k(\mathbf{x}_p, \mathbf{x}_q) = \sigma_f^2 \exp\left(-\frac{1}{2\ell^2} \|\mathbf{x}_p - \mathbf{x}_q\|^2\right) + \sigma_n^2$$

Active learning with Gaussian Processes

Adaptive construction of the constitutive database:

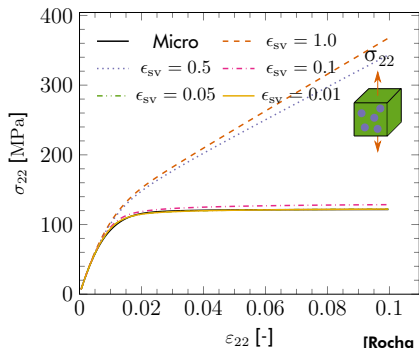
- The model naturally concentrates sampling effort where it is needed



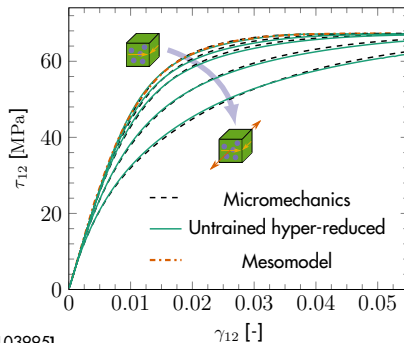
Hyper-reduced models

Beneficial physics-based bias, extrapolates quite well

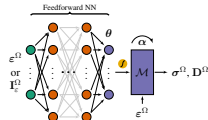
- More modes $\Rightarrow$ increased precision but **lower acceleration**
- Speed-ups are however limited, only about **300 times** for FE² simulations



[Rocha et al. (2020), *Eur J Mech A/Sol*, **82**:103995]

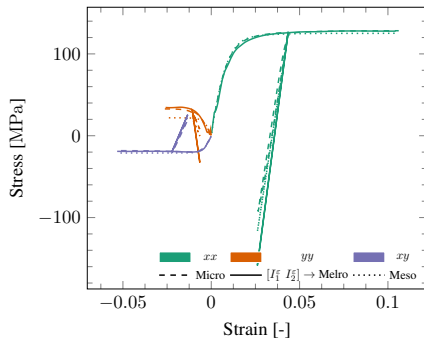
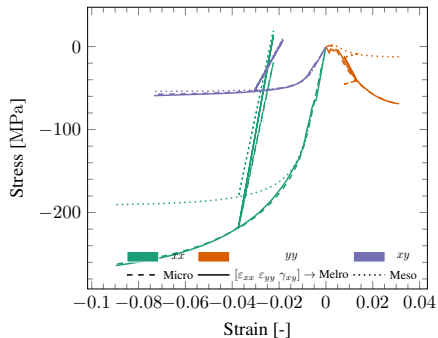


Evolving material properties

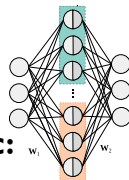


Training only on monotonic paths, predicting unloading/reloading:

- Learns with less data through the use of only two invariants as features
- Can capture path dependency without ever seeing it during training



Physically Recurrent Neural Networks



RNN trained on loading/unloading paths, PRNN only on monotonic:

- RNN needs significantly more data, and with the right patterns

